Research design pitfalls: Key Challenges in Research Design for Business Studies

Authored by Carsten Bergenholtz assisted by ChatGPT-5.2 Pro (15.01.25), based on prior lectures and related documents. Note, you should read Bergenholtz 2026d on Research Designs before this document.

Introduction: The importance of research design

"*The practical problem confronting researchers as they design studies is that (a) there are no hard and fast rules to apply; matching research design to research questions is as much art as science*" (Bono and McNamara 2011: p. 657).

Engaging in high-quality research is a complex and demanding challenge, often underestimated in terms of both the time and resources it requires. Consider that a typical PhD student writes 'just' three articles during a multi-year PhD process. The research design phase is particularly challenging. Choices made early in the study strongly influence the results, and therefore the study's credibility and usefulness (Angrist and Pischke 2010). This document builds on the concepts introduced in Bergenholtz 2026d (Research Designs) and discusses key challenges researchers and students face during the research design process.

The guidance provided here is not about the specific statistical or qualitative techniques for analyzing data, nor about how to theorize from findings. Rather, it focuses on helping students make good research design decisions early on, such as aligning data collection choices with research questions, selecting relevant samples, and defining constructs clearly. A good design is essential because fundamental decisions are hard to revise once data collection has started. If you omit an important variable, it may be difficult (or impossible) to collect it later. If you choose the wrong sample, changing the data-collection method may not solve the problem.

These considerations are not only essential for those doing research but also for those reading and evaluating the work of others. Understanding research design principles enables you to evaluate the choices made in studies, identifying potential weaknesses in the design. This is useful both when assessing information you read in the media or in scientific texts; you want to be able to critically assess whether one should trust the information provided.

The following considerations apply broadly to both qualitative and quantitative studies, though concepts like omitted variable bias and common methods variance are typically associated with quantitative research.

Research questions

A research question guides a study's design, methods, and analysis by clearly defining what the researcher aims to investigate. Well-crafted research questions are essential for making informed research design decisions. Broad or vague questions can derail a study's focus and make findings difficult to interpret. For example, "What influences employee motivation?" lacks the specificity needed for meaningful research. A more focused question like "How do flexible work arrangements impact employee motivation in the tech industry?" provides clear direction and scope.

In qualitative research, precision is particularly important. Rather than asking "How do employees feel about their jobs?"—which could generate overwhelming, unfocused data—researchers might ask "How do employees in high-stress roles cope with new work-life balance policies?" This specificity helps guide data collection and analysis toward meaningful insights.

Ultimately, creating a focused and answerable research question is a vital antecedent to developing a robust research design. When students take the time to refine their research questions early on, they establish a solid foundation for their thesis, ensuring that each design decision supports a coherent, targeted investigation.

Inappropriate samples

Choosing the right sample is crucial for any research study, as it directly influences how well the findings can generalize to a broader context. One frequent mistake researchers make is using samples that do not align with their research question, making it difficult to reach any conclusions. This issue often arises when researchers rely on convenience samples—groups that are easy to access but may not represent the actual population the study aims to explore.

For example, imagine a study designed to understand how employees manage high-stress situations on the job. If this study is done in a laboratory setting using regular students as participants, the results may be limited in their relevance. Students in a lab lack the same pressures, responsibilities, and day-to-day experiences of employees facing real workplace challenges. While students might show certain stress responses in a simulated lab task, their reactions may not accurately reflect how an (experienced) employee would handle high-stress scenarios in a difficult work environment. Consequently, findings from this sample may have limited external & ecological validity for real employees in real workplaces. However, imagine one asks the students to engage in scenarios where they have to consider how they would react to certain kinds of managers in stressful situations, then the sample might be meaningful (cf. Bono & McNamara 2011: p. 658).

A similar issue can arise when researchers select samples from a specific setting or demographic without considering broader usefulness. Suppose a study aims to investigate how employees in various industries respond to changes in team structures but mostly samples employees from startups. Employees in startups might be more adaptable to change compared to those in larger, more hierarchical organizations, where team structures are often stricter. Findings developed from startup employees might therefore fail to represent how employees in different organizational settings respond to similar changes.

Selecting an appropriate sample means choosing participants who closely match the target population or research question. When this alignment is overlooked, the study risks producing results that are limited, which could undermine both its validity and its relevance. By carefully considering the relevance of their sample, researchers can ensure that their findings are not only methodologically correct but also applicable to the real-world population they aim to understand.

Cross-sectional data and change

One of the major limitations in research design arises when studies rely on cross-sectional data – data collected at a single point in time (cf. Bergenholtz 2026d). Cross-sectional studies are often simple and efficient to conduct, yet they generally face serious limitations when it comes to investigating questions of causality and change; which is often the aim of research. When data are gathered only once, it becomes difficult to establish causal relationships between variables with confidence (cf. Isager 2023). For instance, if a study finds a link between employee motivation and productivity, cross-sectional data alone cannot definitively establish whether higher motivation leads to increased productivity or if, conversely, productivity boosts motivation. Without multiple time points or an experimental approach, cross-sectional data lacks the depth needed to identify cause-and-effect relationships.

The limitations of cross-sectional designs not only relate to reverse causality (as outlined above) but also links to the following point about omitted variable bias - which can be an issue in other kinds of research designs as well (e.g. longitudinal designs). As outlined earlier, developing an appropriate research question (one dealing with change or not, for example), therefore, helps clarify whether a cross-sectional design is sufficient or if different types of research designs and types of data are needed.

Omitted variable bias (and how to statistically control for them)

Another common pitfall in research design is omitted variable bias, which occurs when a study fails to account for relevant factors that influence the relationship between variables (i.e., something is 'omitted'). When a critical variable is missing from the analysis, the results may give an incomplete or misleading picture of the relationships. This happens because omitted variables—often confounders that affect both the independent and dependent variables—can distort the observed correlations, creating an illusion of causation where it might not exist (cf. Isager 2023). In other words, omitted variable bias can make it look like two variables are directly related, while in reality, a third factor may be influencing both variables and creating the observed relationship.

For example, **imagine a study finds that people who exercise more report higher job satisfaction. But health can act as a confounder**: healthy people tend to exercise more, and they also tend to be more satisfied at work in general. Hence, health might drive the association between exercise and job satisfaction. If one draws a regression line between exercise and job satisfaction, it might appear to show a clear relationship, but what it actually shows is that unhealthy people with low satisfaction exercise little, while healthy people have high satisfaction and exercise a lot. In order to address this problem, one can in a statistical analysis control for health (see this link for an interactive tool that illustrates this: https://carstenbergenholtz.github.io/Phil-of-Science-tools/statistical_control_intuition.html). By **controlling** for health, we stop comparing "apples and oranges", and aim to compare individuals with similar health status. Practically, you can think of it as doing the comparison "within similar people" (only those with a certain low health level) rather than across very different people (both low and high health people). In this case, we ask: among people with similar baseline health, does more exercise still go together with higher job satisfaction? This reduces plausible alternative explanations, though it should also be clear that this strategy of 'trying' to compare people that are similar is not as effective as randomizing individuals to ensure that individuals in each condition are as similar as possible.

However, adding controls is not simply a case of "more is better." If you add many variables that are only weakly related (or not related) to the story you are testing, the analysis becomes harder to understand, and the main point is hidden or distorted. A better approach is to include a small set of controls that you can justify with common sense: variables that are plausible confounders for your specific relationship (here: baseline health).

Measurement validity

A construct is an abstract concept or theoretical idea that is specifically defined and measured within a study, representing complex phenomena such as “motivation,” “leadership,” or “job satisfaction”. Often such constructs can be difficult to measure directly because they represent broad, intangible ideas that different people might have different understandings of. The task of creating such construct is to ensure that these abstract concepts are translated into specific, measurable variables that accurately capture the phenomenon in question (= high measurement validity).

The process of measuring such constructs goes beyond simply naming an idea; it requires a careful alignment between the theoretical aim of the research and measurement. For example, consider someone interested in measuring firm performance. Firm performance is a broad idea, but it could be measured in various ways: through short-term metrics like quarterly revenue or long-term indicators like market growth. The choice of measurement should align with the specific focus of the study. If the study aims to investigate factors influencing a firm’s immediate performance, then quarterly revenue might be a suitable measure. However, if the goal is to assess sustained performance over time, a single-quarterly revenue measure would be inadequate.

A well-designed construct thus requires a solid measurement approach. Researchers may use existing scales, which are validated tools developed in prior studies, to measure constructs reliably. However, if existing scales are adapted or if new scales are created, researchers must provide evidence that these scales accurately capture the construct. For instance, if a researcher adapts a standard survey on “individual job performance” to measure “team performance,” they must demonstrate that the changed scale is still a reliable and valid measure of the intended construct.

Ensuring measurement validity, therefore, is not only about defining the concept but also about selecting or developing measures that really capture its essence. Misalignment between theory and measurement can weaken the study’s validity, leading to findings that may not accurately represent the theoretical concepts the research set out to investigate.

Common methods variance and overreliance on single-source data

Common methods variance, or common methods bias, occurs when data for both the independent and dependent variables are collected from the same source or using the same method, often through self-reports or surveys. This overlap introduces a risk of systematic bias, where tendencies in the source—like a person’s mood or desire to present themselves positively—impacts both variables in similar ways. This can create an artificial correlation between variables, making it seem like there is a stronger (or weaker) relationship than actually exists.

This is a bit abstract, so let us consider an example: Imagine a study that examines the link between employees’ job satisfaction and their self-reported productivity levels. Both variables are collected through a single survey. Here, if an employee has a generally positive outlook, they might rate both their satisfaction and productivity higher, simply reflecting their general mood rather than a real link between job satisfaction and productivity. And, vice versa, someone with a general negative outlook, or having a bad day, might rate both their satisfaction and productivity lower. When all data come from a single source, like self-reports or a single survey respondent, it risks reflecting more about the source’s biases or perceptions than about objective reality. For example, if managers are asked to rate both team morale and team performance, their personal biases or management style might influence both ratings, producing a misleadingly strong correlation between the two variables.

To minimize common methods bias, researchers can collect independent and dependent variables from different sources (a form of triangulation) or use a mix of methods. In the job satisfaction example, job satisfaction could be gathered through a survey, while productivity could be measured using objective performance metrics. Or, one could ask the employees about their job satisfaction, and ask managers for ratings on the employees' job performance. Such approaches reduce the risk of single-source bias, providing a more accurate picture of the relationships being studied and making the findings more reliable and meaningful.

Summing up

A strong research design is critical for producing credible findings, both for researchers or students producing a thesis. Avoiding common pitfalls—such as relying solely on cross-sectional data, overlooking important variables, using inappropriate samples, falling into common methods bias, or using poorly defined constructs—can significantly improve the quality of your work.

References

Angrist, Joshua D., and Jörn-Steffen Pischke. "The credibility revolution in empirical economics: How better research design is taking the con out of econometrics." *Journal of Economic Perspectives* 24.2 (2010): 3-30.
Bono, Joyce E., and Gerry McNamara. "Publishing in AMJ—part 2: Research design." *Academy of Management Journal* 54.4 (2011): 657-660.
Isager, P. "Why does correlation not equal causation?" (2023) https://pedermisager.org/blog/why_does_correlation_not_equal_causation/ (curriculum)

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